

Transformations of Quadratics in Vertex Form

Comprehensive Notes and Practice

1 Core Concept: Vertex Form

The **vertex form** of a quadratic function is:

$$y = a(x - h)^2 + k \quad (1)$$

Where:

- (h, k) is the **vertex** of the parabola
- a controls the **direction** and **width** of the parabola

2 Transformation Examples

Worked Examples	
Example 1: $y = 2(x - 3)^2 + 1$	Example 2: $y = -\frac{1}{2}(x + 4)^2 - 2$
Starting from $y = x^2$:	Starting from $y = x^2$:
1. Vertical stretch by 2 ($a = 2$) → Narrower parabola	1. Vertical compression by $\frac{1}{2}$ ($ a = \frac{1}{2}$) → Wider parabola
2. Shift right 3 units ($h = 3$) → Vertex: $(0, 0) \rightarrow (3, 0)$	2. Reflection over x-axis ($a < 0$) → Opens downward
3. Shift up 1 unit ($k = 1$) → Vertex: $(3, 0) \rightarrow (3, 1)$	3. Shift left 4 units ($h = -4$) → Vertex: $(0, 0) \rightarrow (-4, 0)$
	4. Shift down 2 units ($k = -2$) → Vertex: $(-4, 0) \rightarrow (-4, -2)$
Final vertex: $(3, 1)$ Opens: Upward ($a > 0$)	Final vertex: $(-4, -2)$ Opens: Downward ($a < 0$)

Transformation	Effect	Example
$a > 1$	Narrower (vertical stretch)	$y = 3x^2$
$0 < a < 1$	Wider (vertical compression)	$y = 0.25x^2$
$a < 0$	Opens down (reflection)	$y = -x^2$
$(x - h), h > 0$	Shift right h units	$y = (x - 5)^2$
$(x + h), h > 0$	Shift left h units	$y = (x + 5)^2$
$+k$	Shift up k units	$y = x^2 + 7$
$-k$	Shift down k units	$y = x^2 - 3$

3 Practice Problems

3.1 Section A: Identifying Transformations

For each function, identify the vertex, direction (up/down), and describe all transformations from $y = x^2$.

1. $y = 3(x - 2)^2 + 5$

Answer:

Vertex: _____ Opens: _____

2. $y = -(x + 1)^2 - 4$

Answer:

Vertex: _____ Opens: _____

3. $y = \frac{1}{2}(x - 6)^2$

Answer:

Vertex: _____ Opens: _____

4. $y = -2(x + 3)^2 + 1$

Answer:

Vertex: _____ Opens: _____

5. $y = (x + 5)^2 - 7$

Answer:

Vertex: _____ Opens: _____

6. $y = 4(x - 1)^2 - 2$

Answer:

Vertex: _____ Opens: _____

7. $y = -\frac{1}{4}(x + 2)^2 + 3$

Answer:

Vertex: _____ Opens: _____

8. $y = -(x - 4)^2$

Answer:

Vertex: _____ Opens: _____

3.2 Section B: Writing Equations from Descriptions

Write the vertex form equation for each description.

9. Vertex at
- $(2, 3)$
- , opens upward, vertical stretch by factor of 2

Answer:

Answer: _____

10. Vertex at
- $(-1, -5)$
- , opens downward, no vertical stretch/compression

Answer:

Answer: _____

11. Vertex at $(0, 4)$, opens upward, vertical compression by factor of $\frac{1}{3}$

Answer:

Answer: _____

12. Vertex at $(7, -2)$, opens downward, vertical stretch by factor of 5

Answer:

Answer: _____

13. The parent function $y = x^2$ shifted left 3 units and down 6 units

Answer:

Answer: _____

3.3 Section D: Graphing Practice

Sketch graphs for these functions. Label the vertex and at least 2 other points.

14. $y = (x - 2)^2 + 1$

15. $y = -\frac{1}{2}(x + 1)^2 + 3$

16. $y = 2(x - 3)^2 - 4$

17. $y = -(x + 2)^2 - 1$

3.4 Section E: Advanced Application

18. A parabola has vertex $(-2, 5)$ and passes through the point $(0, 1)$. Write its equation in vertex form.

Answer:

19. Write an equation for a parabola that opens downward, is wider than $y = x^2$, and has its vertex at $(3, -2)$.

5cm

Answer:

20. If the vertex of $y = a(x - 2)^2 + 3$ is at the point $(2, 3)$, and the parabola passes through $(4, 7)$, find the value of a .

Answer: $a =$

21. A parabola in vertex form $y = a(x - 1)^2 - 8$ passes through the point $(3, 4)$. Find the value of a .

Answer: $a =$

22. The parabola $y = a(x + 5)^2 + k$ has a vertex at $(-5, 7)$ and passes through the point $(-3, 3)$. Find the values of a and k .

Answer

$a =$ _____ $k =$ _____

4 Answer Key

4.1 Section A

1. **Vertex:** $(2, 5)$, Opens up
Vertical stretch by 3, right 2, up 5
2. **Vertex:** $(-1, -4)$, Opens down
Reflection over x -axis, left 1, down 4
3. **Vertex:** $(6, 0)$, Opens up
Vertical compression by $\frac{1}{2}$, right 6
4. **Vertex:** $(-3, 1)$, Opens down
Vertical stretch by 2, reflection, left 3, up 1
5. **Vertex:** $(-5, -7)$, Opens up
Left 5, down 7
6. **Vertex:** $(1, -2)$, Opens up
Vertical stretch by 4, right 1, down 2
7. **Vertex:** $(-2, 3)$, Opens down
Vertical compression by $\frac{1}{4}$, reflection, left 2, up 3
8. **Vertex:** $(4, 0)$, Opens down
Reflection over x -axis, right 4

4.2 Section B

9. $y = 2(x - 2)^2 + 3$
10. $y = -(x + 1)^2 - 5$
11. $y = \frac{1}{3}x^2 + 4$
12. $y = -5(x - 7)^2 - 2$
13. $y = (x + 3)^2 - 6$
14. $y = -(x - 1)^2$

4.3 Section C

15. $y = 5(x - 1)^2$ is narrower ($|a| = 5 > 2$)
16. $y = -3(x + 2)^2 + 1$ opens downward ($a < 0$)
17. $y = (x + 2)^2$ has vertex at $(-2, 0)$, which is further left than $(3, 0)$
18. $y = (x - 1)^2 + 4$ has vertex at height 4, which is higher than height 2

4.4 Section D

19. Vertex: $(2, 1)$, passes through $(0, 5)$, $(1, 2)$, $(3, 2)$, $(4, 5)$
20. Vertex: $(-1, 3)$, passes through $(-3, 1)$, $(0, 2.5)$, $(1, 1)$
21. Vertex: $(3, -4)$, passes through $(2, -2)$, $(4, -2)$, $(5, 4)$
22. Vertex: $(-2, -1)$, passes through $(-3, -2)$, $(-1, -2)$, $(0, -5)$

4.5 Section E

23. Using vertex form: $y = a(x + 2)^2 + 5$

$$\text{Substitute } (0, 1) : \quad 1 = a(0 + 2)^2 + 5$$

$$1 = 4a + 5$$

$$-4 = 4a$$

$$a = -1$$

Answer: $y = -(x + 2)^2 + 5$

24. Sample answer: $y = -\frac{1}{2}(x - 3)^2 - 2$
(Any $|a| < 1$ with negative sign works)

25. Transformations in order:

(a) Vertical stretch by factor of 3

(b) Reflection over x -axis

(c) Shift left 4 units

(d) Shift down 1 unit

26. Using $y = a(x - 2)^2 + 3$

$$\text{Substitute } (4, 7) : \quad 7 = a(4 - 2)^2 + 3$$

$$7 = 4a + 3$$

$$4 = 4a$$

$$a = 1$$

Answer: $a = 1$

27. Using $y = a(x - 1)^2 - 8$

$$\text{Substitute } (3, 4) : \quad 4 = a(3 - 1)^2 - 8$$

$$4 = a(2)^2 - 8$$

$$4 = 4a - 8$$

$$12 = 4a$$

$$a = 3$$

Answer: $a = 3$, so $y = 3(x - 1)^2 - 8$

28. The vertex is $(-5, 7)$, so $k = 7$.

Using $y = a(x + 5)^2 + 7$

$$\text{Substitute } (-3, 3) : 3 = a(-3 + 5)^2 + 7$$

$$3 = a(2)^2 + 7$$

$$3 = 4a + 7$$

$$-4 = 4a$$

$$a = -1$$

Answer: $a = -1, k = 7$

29. Using vertex $(4, -1)$: $y = a(x - 4)^2 - 1$

$$\text{Substitute } (6, 7) : 7 = a(6 - 4)^2 - 1$$

$$7 = a(2)^2 - 1$$

$$7 = 4a - 1$$

$$8 = 4a$$

$$a = 2$$

Answer: $y = 2(x - 4)^2 - 1$

5 Tips for Success

- ✓ **Remember the sign trick:** $(x - h)$ means shift RIGHT, $(x + h)$ means shift LEFT
- ✓ **The vertex is always (h, k) :** Read it directly from the equation
- ✓ **Check the “ a ” value first:** It tells you direction and width
- ✓ **Practice visualizing:** Sketch a quick graph to verify your transformations
- ✓ **Use test points:** Pick easy x -values near the vertex to plot additional points

6 Common Mistakes to Avoid

- × Confusing the sign of h (horizontal shift)
- × Forgetting that $|a| > 1$ makes the graph narrower, not wider
- × Missing the negative sign when $a < 0$
- × Mixing up horizontal and vertical shifts
- × Forgetting to apply all transformations when graphing

Good luck with your practice! Remember: vertex form makes transformations easy to see!